

Secure Dashboards with Traefik Authelia

<https://gemini.google.com/app/b4ec6f41ccb8b50b>

User prompt: i want to add details such as server dashboard , routers info and relevant things using traefik authelia

Response: To route and protect your **Traefik Dashboard**, **Server Dashboard** (e.g., Homepage, Dashy, Heimdall), and **Router/Network Web UIs** using Traefik and Authelia, you need three key components:

1. **Authelia ForwardAuth Middleware** defined in Traefik.
2. **Docker Labels or Dynamic Configuration** for each service/router.
3. **Access Control Policies** configured inside Authelia (configuration.yml).

1. Define the Authelia ForwardAuth Middleware

In your Docker Compose file where Authelia runs, define the reusable authelia ForwardAuth middleware using Docker labels:

```
services:
  authelia:
    image: authelia/authelia:latest
    container_name: authelia
    restart: unless-stopped
    volumes:
      - ./authelia/config:/config
    networks:
      - proxy
    labels:
      - "traefik.enable=true"
      # Authelia Web Portal Route
      - "traefik.http.routers.authelia.rule=Host(`auth.example.com`)"
      - "traefik.http.routers.authelia.entrypoints=websecure"
      - "traefik.http.routers.authelia.tls=true"
      - "traefik.http.routers.authelia.tls.certresolver=myresolver"

      # Core ForwardAuth Middleware (used by other apps)
      - "traefik.http.middlewares.authelia.forwardauth.address=http://authelia:9091/api/authz/forward-auth"
      - "traefik.http.middlewares.authelia.forwardauth.trustForwardHeader=true"
      - "traefik.http.middlewares.authelia.forwardauth.authResponseHeaders=Remote-User,Remote-Groups,Remote-Name"
```

2. Expose and Secure Traefik's Own Dashboard

Traefik provides an internal API/Dashboard (api@internal). You can attach Docker labels to the Traefik container to expose and protect it:

```
services:
  traefik:
    image: traefik:v3.1
    container_name: traefik
    restart: unless-stopped
    command:
      - "--api.dashboard=true" # Enable dashboard
      - "--providers.docker=true"
      - "--providers.docker.exposedbydefault=false"
      - "--entrypoints.websecure.address=:443"
    ports:
      - "80:80"
      - "443:443"
    volumes:
      - /var/run/docker.sock:/var/run/docker.sock:ro
      - ./traefik/config/dynamic:/etc/traefik/dynamic # Dynamic config folder for routers
    networks:
      - proxy
    labels:
      - "traefik.enable=true"
      # Traefik Dashboard Route
      - "traefik.http.routers.traefik-dashboard.rule=Host(`traefik.example.com`)"
      - "traefik.http.routers.traefik-dashboard.entrypoints=websecure"
      - "traefik.http.routers.traefik-dashboard.service=api@internal"
      - "traefik.http.routers.traefik-dashboard.tls=true"
      - "traefik.http.routers.traefik-dashboard.tls.certresolver=myresolver"
      # Protect with Authelia middleware defined on the authelia service
      - "traefik.http.routers.traefik-dashboard.middlewares=authelia@docker"
```

3. Add a Server Dashboard (e.g., Homepage)

If you are running a server dashboard container on the same Docker network, attach the `authelia@docker` middleware directly in its labels:

```
services:
  homepage:
    image: ghcr.io/gethomepage/homepage:latest
    container_name: homepage
    restart: unless-stopped
    networks:
      - proxy
    labels:
      - "traefik.enable=true"
      - "traefik.http.routers.homepage.rule=Host(`dash.example.com`)"
      - "traefik.http.routers.homepage.entrypoints=websecure"
      - "traefik.http.routers.homepage.tls=true"
      - "traefik.http.routers.homepage.tls.certresolver=myresolver"
      # Attach Authelia authentication
      - "traefik.http.routers.homepage.middlewares=authelia@docker"
      - "traefik.http.services.homepage.loadbalancer.server.port=3000"
```

4. Route External Hardware / Routers (OPNsense, pfSense, OpenWrt)

For physical hardware like network routers or switches running outside Docker, use Traefik's **YAML dynamic file provider**.

Create `./traefik/config/dynamic/external_routers.yml`:

```
http:
  routers:
    router-gui:
      rule: "Host(`router.example.com`)"
      entryPoints:
        - "websecure"
      tls:
        certResolver: "myresolver"
        service: "router-service"
      # Apply Authelia middleware created via Docker
      middlewares:
        - "authelia@docker"

  services:
    router-service:
      loadBalancer:
        servers:
          - url: "https://192.168.1.1" # IP address of your physical router UI
```

5. Configure Authelia Access Control Rules

In Authelia's `configuration.yml`, define access requirements for each subdomain:

```
server:
  host: 0.0.0.0
  port: 9091

session:
  domain: example.com

access_control:
  default_policy: deny
  rules:
    # 1. Allow public access to Authelia's portal
    - domain: "auth.example.com"
      policy: bypass

    # 2. Require Two-Factor Auth for Traefik Dashboard and Router UI
    - domain:
      - "traefik.example.com"
      - "router.example.com"
      policy: two_factor

    # 3. Require Single-Factor Auth for Main Server Dashboard
    - domain: "dash.example.com"
      policy: one_factor
```

Verification Checklist

1. **DNS:** Ensure wildcard or subdomains (auth, traefik, dash, router) point to your Traefik host IP.
2. **Network:** Ensure traefik, authelia, and dashboard containers share the same Docker network (proxy).
3. **Test Access:** Navigating to `traefik.example.com` should automatically redirect to `auth.example.com` for authentication before returning to the dashboard.

Sources

1. [Authelia + Traefik Setup Guide | Blog](#)
 2. [Traefik SSO with Authelia across multiple hosts - Florian Jensen's Blog](#)
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